

Geometry & Coordinate Geometry

ANSWER KEY



Distance and midpoint

- 1 Find the distance between $(1, 2)$ and $(4, 6)$.
 $d = \sqrt{(4 - 1)^2 + (6 - 2)^2} = \sqrt{9 + 16} = \sqrt{25} = 5.$
 - 2 Find the midpoint of $(2, -3)$ and $(8, 5)$.
Midpoint $= \left(\frac{2 + 8}{2}, \frac{-3 + 5}{2} \right) = (5, 1).$
 - 3 Find the distance between $(-2, 1)$ and $(3, -11)$.
 $d = \sqrt{(3 - (-2))^2 + (-11 - 1)^2} = \sqrt{25 + 144} = \sqrt{169} = 13.$
 - 4 Find the midpoint of $(-4, 7)$ and $(6, -3)$.
Midpoint $= \left(\frac{-4 + 6}{2}, \frac{7 - 3}{2} \right) = (1, 2).$
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Gradient (slope)

- 5 Find the gradient of the line through $(1, 2)$ and $(5, 10)$.
 $m = \frac{10 - 2}{5 - 1} = \frac{8}{4} = 2.$
 - 6 Find the gradient of the line through $(-3, 4)$ and $(2, -6)$.
 $m = \frac{-6 - 4}{2 - (-3)} = \frac{-10}{5} = -2.$
 - 7 Find the gradient of the line through $(0, 0)$ and $(6, 9)$.
 $m = \frac{9 - 0}{6 - 0} = \frac{9}{6} = \frac{3}{2}.$
 - 8 State whether the line through $(1, 1)$ and $(4, 1)$ is horizontal, vertical, or neither.
Both points have $y = 1$, so the line is horizontal (gradient $= 0$).
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Equations of lines

- 9 Find the equation of the line with gradient 2 passing through $(3, 5)$, in the form $y = mx + c$.
 $y - 5 = 2(x - 3) \Rightarrow y = 2x - 1.$
- 10 Find the equation of the line through $(0, 4)$ and $(2, 10)$.
 $m = \frac{10 - 4}{2 - 0} = 3.$ Using the y -intercept 4: $y = 3x + 4.$
- 11 Find the x - and y -intercepts of the line $2x + 3y = 12$.
 x -intercept ($y = 0$): $x = 6.$ y -intercept ($x = 0$): $y = 4.$

- 12 Write the equation of the line parallel to $y = 4x - 1$ passing through $(2, 9)$.
Same gradient 4: $y - 9 = 4(x - 2) \Rightarrow y = 4x + 1$.
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Parallel and perpendicular lines

- 13 State the gradient of a line perpendicular to a line with gradient 5.
 $m = -\frac{1}{5}$.
- 14 Determine whether the lines $y = 2x + 3$ and $y = 2x - 7$ are parallel, perpendicular, or neither.
Parallel — both have gradient 2.
- 15 Find the equation of the line perpendicular to $y = -\frac{1}{3}x + 2$, passing through $(3, 4)$.
The perpendicular gradient is 3. $y - 4 = 3(x - 3) \Rightarrow y = 3x - 5$.
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The equation of a circle

- 16 Write the equation of a circle centered at the origin with radius 6.
 $x^2 + y^2 = 36$.
- 17 Write the equation of a circle centered at $(2, -3)$ with radius 5.
 $(x - 2)^2 + (y + 3)^2 = 25$.
- 18 Find the radius and center of the circle $(x - 1)^2 + (y + 4)^2 = 49$.
Center: $(1, -4)$. Radius: 7.
- 19 Find the equation of the circle centered at the origin that passes through $(3, 4)$.
Radius $= \sqrt{3^2 + 4^2} = 5$. Equation: $x^2 + y^2 = 25$.
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Circles: chords and arcs

- 20 Points A and B lie on a circle of radius 5 centered at O , with central angle $\angle AOB = 90^\circ$. Find the length of chord AB , using $AB = 2r \sin\left(\frac{\theta}{2}\right)$.
 $AB = 2(5)\sin(45^\circ) = 10(0.7071) \approx 7.07$.
- 21 A chord of a circle with radius 10 cm is 12 cm long. Find the distance from the center to the chord.
Half the chord is 6 cm. Distance $= \sqrt{10^2 - 6^2} = \sqrt{100 - 36} = \sqrt{64} = 8$ cm.
- 22 Find the circumference of a circle with radius 9 cm, in terms of π .
 $C = 2\pi r = 18\pi$ cm.
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Polygon angle sums

- 23 Find the sum of the interior angles of a hexagon (6 sides).
 $(n - 2)180^\circ = (6 - 2)180^\circ = 720^\circ$.
- 24 Find the measure of each interior angle of a regular pentagon.
Sum $= (5 - 2)180^\circ = 540^\circ$. Each angle $= \frac{540^\circ}{5} = 108^\circ$.
- 25 A polygon has an interior angle sum of 1440° . Find the number of sides.
 $(n - 2)180^\circ = 1440^\circ \Rightarrow n - 2 = 8 \Rightarrow n = 10$.
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Perimeter and area

- 26 A rectangle has length 12 and width 7. Find its perimeter and area.
Perimeter $= 2(12 + 7) = 38$. Area $= 12 \times 7 = 84$.
- 27 A right triangle has legs 6 and 8. Find its area and hypotenuse.
Area $= \frac{1}{2}(6)(8) = 24$. Hypotenuse $= \sqrt{36 + 64} = \sqrt{100} = 10$.
- 28 A regular hexagon has side length 4, with area $A = \frac{3\sqrt{3}}{2}s^2$. Find its area in exact form.
 $A = \frac{3\sqrt{3}}{2}(16) = 24\sqrt{3}$.
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Coordinate geometry: identifying shapes

- 29 Show that the triangle with vertices $A(0, 0)$, $B(4, 0)$, $C(4, 3)$ is a right triangle.
 AB is horizontal and BC is vertical, so they are perpendicular; the right angle is at B .
- 30 Determine whether the points $A(-2, 1)$, $B(2, 4)$, $C(5, 0)$ form a right triangle, using the squared side lengths.
 $AB^2 = 4^2 + 3^2 = 25$. $BC^2 = 3^2 + 4^2 = 25$. $AC^2 = 7^2 + 1^2 = 50$. Since $AB^2 + BC^2 = 50 = AC^2$, the triangle has a right angle at B .
- 31 Determine whether the triangle with vertices $A(1, 2)$, $B(5, 2)$, $C(3, 6)$ is isosceles.
 $AC = \sqrt{2^2 + 4^2} = \sqrt{20}$ and $BC = \sqrt{2^2 + 4^2} = \sqrt{20}$, so $AC = BC$ and the triangle is isosceles.
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The perpendicular bisector of a segment

- 32 Find the equation of the perpendicular bisector of the segment joining $(2, 3)$ and $(8, 7)$.
Midpoint $= (5, 5)$. Segment gradient $= \frac{7 - 3}{8 - 2} = \frac{2}{3}$, so the perpendicular gradient is $-\frac{3}{2}$. Equation:
 $y - 5 = -\frac{3}{2}(x - 5) \Rightarrow y = -\frac{3}{2}x + \frac{25}{2}$.

- 33** Find the equation of the perpendicular bisector of the segment joining $(0, 0)$ and $(6, 8)$.
 Midpoint $= (3, 4)$. Segment gradient $= \frac{8}{6} = \frac{4}{3}$, so the perpendicular gradient is $-\frac{3}{4}$. Equation:
 $y - 4 = -\frac{3}{4}(x - 3) \Rightarrow y = -\frac{3}{4}x + \frac{25}{4}$.
- 34** A line segment has endpoints $(1, 5)$ and $(7, 5)$. Find the equation of its perpendicular bisector.
 Midpoint $= (4, 5)$. The segment is horizontal (gradient 0), so the perpendicular bisector is the vertical line $x = 4$.

Word problems: geometry on a coordinate grid

- 35** This question has two parts. A city park is designed on a coordinate grid (in meters), with two gates at $A(0, 0)$ and $B(80, 60)$.
- a** Find the straight-line distance between the two gates.
 $AB = \sqrt{80^2 + 60^2} = \sqrt{6400 + 3600} = \sqrt{10000} = 100$ meters.
- b** A fountain is placed at the midpoint between the two gates. Find its coordinates. Midpoint
 $= \left(\frac{0+80}{2}, \frac{0+60}{2} \right) = (40, 30)$.
- 36** This question has two parts. A circular irrigation sprinkler is centered at $(5, 5)$ on a farm's coordinate grid (in meters) and has a radius of 12 meters.
- a** Write the equation of the circle covered by the sprinkler. $(x - 5)^2 + (y - 5)^2 = 144$.
- b** Determine whether a tree at $(14, 9)$ lies inside, on, or outside the watered region. Distance from center $= \sqrt{(14 - 5)^2 + (9 - 5)^2} = \sqrt{81 + 16} = \sqrt{97} \approx 9.85$ meters, which is less than the radius 12, so the tree lies inside the watered region.